

APPENDIX B - CORE SOLUTIONS

VERIFIED PROOFS

FAIL-CLOSED AUDITS

{{ SOLUTIONS_DIRECTORY: Complete Verified Solutions for Appendix A Practice Items }}

SOLUTION: SOL-PRAC-01

TARGET: PRAC-01

{{ PRIMARY_CAPABILITY: CAP-ASSIGN-OXIDATION-NUMBER-CORE }}

Reasoning Steps:

1. Identify compound type: Phosphoric acid (H_3PO_4) is a neutral molecule (Net charge = 0).
2. Apply Rule Priority 4 (Hydrogen): Hydrogen is bonded to non-metals $\Rightarrow \text{H} = +1$.
Total H contribution = $3 \times (+1) = +3$.
3. Apply Rule Priority 5 (Oxygen): Standard oxide bonding $\Rightarrow \text{O} = -2$. Total O = $4 \times (-2) = -8$.
4. Apply Priority 7 (Sum Rule): Let ON of P = x.
 $(+3) + x + (-8) = 0 \Rightarrow x - 5 = 0 \Rightarrow x = +5$.

{{ FINAL_RESPONSE }}: The oxidation state of Phosphorus in H_3PO_4 is +5.

{{ VERIFICATION_STEP }}: $(+3) + (+5) + (-8) = 0$. Charge balance is exactly satisfied.

{{ CONDITION_NOTE }}: Phosphorus uses all 5 valence electrons in bonding; +5 is its maximum state.

SOLUTION: SOL-PRAC-02

TARGET: PRAC-02

{{ PRIMARY_CAPABILITY: CAP-TRACK-ELEMENT-CHARGE-CHANGE }}

Reasoning Steps:

1. Assign ON to every reactant species:
 - * Free Zinc metal Zn(s): Elemental state $\Rightarrow \text{ON} = 0$.
 - * In $\text{HCl}(\text{aq})$: $\text{H} = +1$, $\text{Cl} = -1$.
2. Assign ON to every product species:
 - * In $\text{ZnCl}_2(\text{aq})$: Group 12 metal Zn = +2; two chloride ions each $\text{Cl} = -1$.
 - * In $\text{H}_2(\text{g})$: Free elemental diatomic hydrogen gas $\Rightarrow \text{ON} = 0$.
3. Track direction of electron transfer:
 - * Zinc: Changes from 0 to +2 $\Rightarrow$ Loss of $2\text{e}^- \Rightarrow$ UNDERGOES OXIDATION.
 - * Hydrogen: Changes from +1 to 0 $\Rightarrow$ Gain of 1e^- per atom $\Rightarrow$ UNDERGOES REDUCTION.
 - * Chloride: Stays at -1 $\Rightarrow$ Spectator ion (unchanged).

{{ FINAL_RESPONSE }}: Zinc is oxidised (0 $\rightarrow$ +2); Hydrogen is reduced (+1 $\rightarrow$ 0).

{{ VERIFICATION_STEP }}: 1 mole Zn loses 2e^- ; 2 moles H^+ gain $2(1\text{e}^-) = 2\text{e}^-$. 100% conserved.

SOLUTION: SOL-PRAC-03

TARGET: PRAC-03

{{ PRIMARY_CAPABILITY: CAP-ATTACH-REDOX-ROLES }}

Reasoning Steps:

1. Identify reacting species and changes in $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$:
 - * Iron in Fe_2O_3 : $2\text{Fe} + 3(-2) = 0 \Rightarrow \text{Fe} = +3$. In product $\text{Fe}(\text{s})$: $\text{Fe} = 0$.
Fe decreases from +3 to 0 $\Rightarrow$ Reduction. Therefore, Fe_2O_3 is the OXIDISING AGENT.
 - * Carbon in CO : $\text{C} + (-2) = 0 \Rightarrow \text{C} = +2$. In product CO_2 : $\text{C} + 2(-2) = 0 \Rightarrow \text{C} = +4$.
C increases from +2 to +4 $\Rightarrow$ Oxidation. Therefore, CO is the REDUCING AGENT.
2. Calculate electron count per formula unit of Fe_2O_3 :
 - Each Iron atom decreases by 3 units: $\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}^0$.
 - 1 formula unit of Fe_2O_3 contains 2 Iron atoms $\Rightarrow 2 \times 3\text{e}^- = 6$ electrons accepted.
 - Check: 3 moles of CO each lose $2\text{e}^- \Rightarrow 3 \times 2\text{e}^- = 6$ electrons donated. (Matches exactly).

{{ FINAL_RESPONSE }}: Oxidising Agent = Fe_2O_3 ; Reducing Agent = CO ; 6 moles e^- transferred.

{{ VERIFICATION_STEP }}: Agent names attached to ENTIRE reactant species, not isolated atoms.

SOLUTION: SOL-PRAC-04

TARGET: PRAC-04

{{ PRIMARY_CAPABILITY: CAP-DISCRIMINATE-PEROXIDE-EXCEPTION }}

Reasoning Steps (Causal Misconception Repair):

1. Physical impossibility check: Barium is an alkaline earth metal (Group 2, $[\text{Xe}] 6\text{s}^2$).
Its ionization energies allow only a +2 oxidation state (loss of two 6s electrons).
 $\text{Ba} = +4$ would require breaking into the stable closed noble-gas shell of Xenon (impossible).
2. Rule Priority Hierarchy: Priority 2 (Group 2 metal is ALWAYS +2) overrides Priority 5 (Oxygen).
Therefore, Ba MUST be assigned +2 fixed.
3. Evaluate Oxygen via Sum Rule:
 - $(+2) + 2(\text{ON}_\text{O}) = 0 \Rightarrow 2(\text{ON}_\text{O}) = -2 \Rightarrow \text{ON of Oxygen} = -1$.
4. Structural Confirmation: BaO_2 contains the peroxide linkage $[\text{O}-\text{O}]^{2-}$.

{{ FINAL_RESPONSE }}: Student claim is FALSE. Ba is fixed at +2; Oxygen is -1 (Peroxide).

{{ CONDITION_NOTE }}: Group 1 and 2 metal fixed rules strictly precede the Oxygen default rule.